Comprehensive Project Scope Document

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1 Introduction

1.1 Background and Context

The project aims to bridge biological and physical models to study the heterogeneity behavior in bioreactors. It involves simulating cell performance inside a reactor with high substrate variability using biological/metabolic modeling. The project is based at the Novo Nordisk Center for Biosustainability at DTU, known for its expertise in industrial biotechnology and the design of cell factories. The Quantitative Modelling of Cell Metabolism group focuses on understanding cellular behavior in different environments through mathematical simulations applied to biology.

1.2 Overview of Objectives and Proposed Methods

The project objectives include analyzing big and complex datasets from advanced biological/CFD simulations, identifying reaction clusters in dynamic metabolic networks, utilizing dimensionality reduction techniques to plot and represent multi-dimensional metabolic networks, and learning to use an HPC cluster for computing complicated tasks.

2 Scope of the Project

2.1 Description of the Project's Focus

The project focuses on studying the behavior of cells inside a bioreactor with high substrate variability in a production industrial environment. It involves the analysis of internal cell data obtained from tracking cells moving inside the reactor for a certain time. The data includes time series readings of substrate concentration seen by the cell at each time point, computation of substrate uptake rate (qS) at each time point, metabolic analysis using dynamic Flux Balance Analysis (dFBA), and the generation of a pd.df with the internal fluxes of the cell at each time point.

2.2 Objective 1: Problems to be Solved

The project aims to address the following problems:

- Identifying relevant information to extract from the internal cell data
- Exploring the possibility of dataset reduction
- Analyzing similarities and trends within a single cell lifeline (e.g., cycles inside the reactor)
- Identifying similarities and trends within lifeline groups (e.g., clusters of cells)
- Determining meaningful metabolic states of the cells (e.g., cycles) and the reactions that define metabolic clusters
- Exploring innovative ways to represent the data

2.3 Importance and Relevance

Understanding the behavior of cells in a bioreactor with substrate variability is crucial for optimizing bioprocesses in industrial settings. By studying the internal cell data and identifying meaningful metabolic states, the project aims to contribute to the development of strategies for improved bioprocess performance and efficiency.

3 Literature Review

3.1 Overview of Relevant Literature

3.1.1 Auto-encoders Based Dimensionality Reduction

The paper titled "Auto-encoder based dimensionality reduction" explores the dimensionality reduction ability of auto-encoder, a three-layered neural network commonly used in deep learning. The authors aim to understand the unique properties of auto-encoder compared to state-of-the-art dimensionality reduction methods and its potential contribution to the success of deep learning.

The study involves experiments conducted on both synthesized and real datasets, including two and three-dimensional spaces for better visualization, as well as the MNIST and Olivetti face datasets. The results demonstrate that auto-encoder can indeed learn distinct representations compared to other methods. It exhibits the capability to capture complex structures, adaptively extract features, compress data, and provide accurate reconstructions. The experiments also investigate the influence of the number of hidden layer nodes on the performance of auto-encoder, revealing a possible relationship between the number of nodes and the intrinsic dimensionality of the input data.

Overall, the findings highlight the effectiveness of auto-encoder in dimensionality reduction and its potential as a building block in deep learning architectures. The unique properties of auto-encoder, such as its ability to detect repetitive structures, make it a valuable tool for various applications. Moreover, the study suggests that choosing the appropriate number of hidden layer nodes can enhance the performance of auto-encoder in reducing dimensionality.

3.1.2 A Folded Neural Network Autoencoder for Dimensionality Reduction

The paper introduces a new structure called the folded autoencoder for dimensionality reduction. It addresses the challenge of handling high-dimensional data by reducing computational costs while improving performance. The folded autoencoder is designed based on the symmetric property of the conventional autoencoder, reducing the number of weights that need to be tuned. The authors demonstrate the effectiveness of the folded autoencoder by comparing it with the conventional autoencoder and principal component analysis (PCA) on the MNIST dataset. The results show that the folded autoencoder outperforms PCA and achieves similar performance to the conventional autoencoder in reconstructing images. Furthermore, the training time of the folded autoencoder is reduced by 17.2

In conclusion, the proposed folded autoencoder structure offers a promising approach to dimensionality reduction. It not only reduces computational costs but also improves the generalization performance compared to conventional approaches such as PCA. The results obtained on the MNIST dataset demonstrate the effectiveness of the folded autoencoder in reconstructing images. Further research directions include theoretical analysis of the framework, evaluation on larger and more complex datasets, and integration with other advanced techniques such as convolutional networks and incremental learning. The folded autoencoder presents a significant contribution to the field of dimensionality reduction and holds potential for various applications in handling high-dimensional data.

3.2 Key Studies and Findings

3.3 Previous Work on Metabolic Analysis

4 Methodology

4.1 Starting Point: From Lifeline to Internal Cell Data

4.2 Objective 1: Data Analysis of the Internal Cell Data

Autoencoders: Autoencoders play a crucial role in this project, as they are neural network models designed for unsupervised learning and dimensionality reduction. They consist of two main components: an encoder and a decoder. Working together, the encoder compresses the input data into a latent space, while the decoder reconstructs it. By minimizing the reconstruction error between the original and reconstructed data, autoencoders capture salient features and extract meaningful representations. The compressed latent space obtained from the autoencoder can be used for subsequent analysis and interpretation. By leveraging the reduced-dimensional representation, it becomes easier to explore patterns, extract meaningful features, and perform various analysis techniques. The dimensionality reduction aspect of autoencoders helps to focus on the essential characteristics of the data, making it more amenable for further analysis tasks such as clustering, visualization, or supervised learning. Objectives 4.2.1 and 4.2.2 focus on building an autoencoder model where as subsequent objecives focus on analysis and use for latent space from the autoencoder models for further processing.

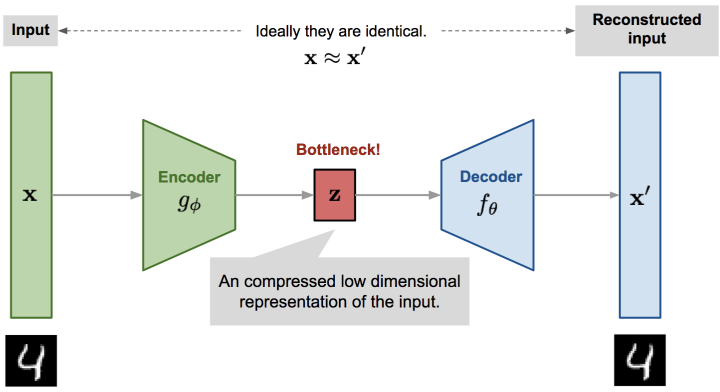

Figure 1: Basic Autoencoder Architecture

4.2.1 Identifying relevant information to extract from the internal cell data

Input data from the domain can then be provided to the autoencoder model, and the output of the model at the bottleneck, which represents the compressed latent space, can be used as a feature vector. This feature vector can be utilized in various ways, such as feeding it into a supervised learning model for classification or regression tasks, enabling visualization of the data in a lower-dimensional space, or more generally, facilitating dimensionality reduction. By leveraging the capabilities of autoencoders, particularly LSTM autoencoders in this project, the goal is to extract the most informative and relevant features that capture the essential characteristics of the internal cell data. This process effectively reduces the complexity of the data while preserving its temporal nature, enabling further analysis and interpretation of the cell behavior within the bioreactor.

4.2.2 Exploring the possibility of dataset reduction

This objective aims to reduce the dimensionality of the internal cell data using autoencoders, specifically leveraging the latent space representation. Autoencoders are neural network models for unsupervised learning and dimensionality reduction. They compress the input into a latent space through an encoder and reconstruct it using a decoder. By minimizing the reconstruction error between the original and reconstructed data, autoencoders capture salient features and extract meaningful representations.

Traditional methods like Principal Component Analysis (PCA) are not well-suited for temporal data due to its temporal dependencies and sequential nature. PCA fails to capture relevant temporal patterns, interactions between reactions, and important information related to temporal dynamics. In contrast, autoencoders are better equipped to preserve temporal dynamics and capture important features. They are specifically designed for time series analysis and feature selection in order to address these limitations.

LSTM (Long Short-Term Memory) autoencoders, a variant of autoencoders utilizing LSTM layers, are particularly appropriate for handling temporal data. LSTM networks are capable of capturing long-term dependencies and temporal relationships in sequential data. In the context of this project, LSTM autoencoders can effectively reduce the number of reaction fluxes while preserving the key temporal dynamics. By leveraging the advantages of LSTM autoencoders, the dataset can be reduced in dimensionality while retaining important information regarding the behavior of cells inside the bioreactor.

The goal of this objective is to employ LSTM autoencoders to reduce the number of reaction fluxes in the internal cell data. By doing so, the project aims to extract the most informative and relevant features, effectively reducing the complexity of the data while preserving its temporal nature. This reduction in dimensionality facilitates subsequent analysis and interpretation of the cell behavior within the bioreactor.

4.2.3 Analyzing similarities and trends within a single cell lifeline (e.g., cycles inside the reactor)

One of the objectives of this project is to analyze the similarities and trends within a single cell lifeline, such as the cycles inside the reactor. To achieve this, the latent space or low-dimensional representation obtained from the autoencoder can be leveraged. The reduced-dimensional representation captures the essential characteristics of the internal cell data while preserving the temporal dynamics.

Clustering Techniques for Lifeline Similarities and Trends To identify similarities and trends within a single cell lifeline, clustering techniques tailored for time series data can be utilized. Methods such as k-means, DBSCAN (Density-Based Spatial Clustering of Applications with Noise), or hierarchical clustering can be applied to the latent space. These techniques group similar data points together, revealing distinct phases or states that a cell goes through during its lifeline. By studying the behavior of cells within each cluster, valuable insights into the temporal dynamics and characteristic patterns exhibited during different stages of the lifeline can be obtained.

Dimensionality Reduction Techniques for Lifeline Visualization Dimensionality reduction techniques designed for time series data, such as t-SNE (t-Distributed Stochastic Neighbor Embedding) or UMAP (Uniform Manifold Approximation and Projection), can be employed to visualize the latent space representation. These methods project the latent space into a lower-dimensional space while preserving the temporal relationships between data points. By visualizing the lifeline in this reduced-dimensional space, meaningful patterns, relationships, and transitions within the lifeline can be revealed.

Machine Learning Models for Analyzing Temporal Dependencies Machine learning models suitable for analyzing temporal dependencies within a single cell lifeline can provide valuable insights. Recurrent neural networks (RNNs) or long short-term memory (LSTM) networks, for example, can be

employed to model the sequential relationships within the lifeline. These models capture long-term dependencies and enable the identification of recurring patterns, cycles, or trends within the lifeline.

By combining the power of autoencoders for dimensionality reduction, clustering algorithms for identifying lifeline phases, dimensionality reduction techniques for visualization, and appropriate machine learning models for analyzing temporal dependencies, this project aims to uncover valuable insights into the behavior of cells within a bioreactor. These insights contribute to a deeper understanding of lifecycles and dynamics, facilitating the optimization and control of bioprocesses in industrial settings.

4.2.4 Identifying similarities and trends within lifeline groups (e.g., clusters of cells)

Another objective of this project is to identify similarities and trends within lifeline groups, which refer to clusters of cells that exhibit similar behavior. By leveraging the latent space or low-dimensional representation obtained from the autoencoder trained on all lifelines, it becomes possible to analyze and characterize these groups.

By applying clustering algorithms, such as k-means or hierarchical clustering, to the latent space, cells with similar behavior can be grouped together. These lifeline clusters represent distinct cohorts of cells that share common metabolic patterns or response mechanisms.

Analyzing the lifeline clusters enables the identification of meaningful trends and similarities within groups of cells. For example, it may reveal specific metabolic states or transitions that are prevalent among certain groups. By examining the temporal evolution of the cells within each cluster, insights into the shared behavior and response patterns can be gained.

Furthermore, the low-dimensional representation in the latent space can be visualized using techniques like scatter plots or heatmaps. Visual exploration allows for a better understanding of the relationships between lifeline groups, highlighting patterns and transitions that are not apparent in the original high-dimensional space. This visualization aids in the interpretation of the behavior of cells within the bioreactor, contributing to the development of strategies for improved bioprocess performance and efficiency.

4.2.5 Determining meaningful metabolic states of the cells (e.g., cycles) and the reactions that define metabolic clusters

4.2.6 Exploring innovative ways to represent the data

4.3 Objective 2: Try More Complex Metabolic Models (Second Phase)

4.4 Innovative Ideas & Strategies

5 References

5.1 Comprehensive List of Cited Literature and References